GREEN VALLEY EDUCATIONAL INSTITUTE (GVEI)

DEPARTMENT OF INFORMATION & COMMUNICATION TECHNOLOGY (ICT)

COMPUTER LABORATORY, INFRASTRUCTURE & ICT INTEGRATION (2026–2027)

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1. Purpose and Strategic Objective

The ICT Department recognises the computer laboratory and digital infrastructure as essential, mission-critical components of effective technology education. Their value lies not merely in the nominal availability of equipment, but in their operational readiness, safe and disciplined utilisation, system reliability, and contribution to meaningful teaching and learning.

This policy framework establishes a structured, evidence-based mechanism for cataloging laboratory resources, executing preventive equipment maintenance, auditing technical issues, tracking scheduled laboratory utilisation, and evaluating the authentic pedagogical integration of ICT into daily classroom instruction.

> **STRATEGIC INFRASTRUCTURE MANDATE**
> *"The strategic objective is to demonstrate that GVEI's ICT infrastructure is managed systematically, technical concerns are documented and resolved transparently, digital resources are deployed purposefully, and continuous infrastructure enhancements are substantiated by authentic evidence."*

2. Computer Laboratory Inventory and Asset Management

The department maintains an updated physical and digital inventory of computers, peripherals, network hardware, and display equipment across all computer laboratories. Each asset is tracked by its unique hardware identifier, designated wing, operational health, and maintenance record.

S. No.	Asset / Equipment Category	Asset ID / Serial Range	Laboratory / Wing	Total Qty	Operational Status	Technical Remarks
1	**Desktop Computers (Core i5/i7)**	GVEI-ICT-PC-001 to 085	Senior Lab, Junior Lab & Girls Lab	90	**Operational 85**	3 units scheduled for RAM upgrade
2	**LED Monitors (21.5-inch)**	GVEI-ICT-MON-001 to 085	Senior Lab , Junior Lab & Girls Lab	90	**Operational 90**	Full functional display verification
3	**Keyboards & Optical Mice**	GVEI-ICT-INP-001 to 090	Senior Lab , Junior Lab & Girls Lab	90 sets	**Operational (90/90)**	3 spare sets kept in lab

						reserve
4	**Network Printers & Scanners**	GVEI-ICT-PRN-01 to 04	ICT Faculty & Senior Lab	0	**0**	LaserJet network shared printers
5	**Networking Equipment (Switches/Routers)**	GVEI-NET-SW-01 to 06	Server Room & Lab Racks	6 units	**Operational (6/6)**	Gigabit managed switches & fiber uplink
6	**Interactive Panels & Ceiling Projectors**	GVEI-ICT-DISP-01 to 05	Computer Labs & AV Room	3	**Operational (3/3)**	HDMI enabled ceiling projection systems
7	**UPS & Power Backup Arrays**	GVEI-PWR-UPS-01 to 03	Main Power Inverter Room	3 banks	**Operational (3/3)**	Online centralized UPS backup system

Verification Mandate: Inventory quantities and working-status metrics must reflect verified physical stock-taking registers rather than unsubstantiated estimates.

3. Hardware Condition and Maintenance Records

Maintaining high hardware availability requires routine preventive servicing, rapid fault reporting, and transparent tracking of all repairs:

Inspection Date	Equipment / Asset ID	Observed Issue / Symptom	Diagnostic Action Taken	Reported To / Technician	Current Status
2026-04-10	**GVEI-ICT-PC-014**	OS boot failure & bad sectors	Drive cloned and replaced with 512GB SSD	Hardware Support Specialist	**Resolved & Verified**
2026-05-18	**GVEI-ICT-PC-042**	Display artifacts / loose VGA cable	Reseated GPU connection & replaced cable	Internal Lab Administrator	**Resolved & Verified**
2026-06-25	**GVEI-NET-SW-02**	Intermittent port packet drop	Port reconfigured and firmware patched	Network Systems Engineer	**Resolved & Verified**
2026-07-14	**GVEI-ICT-DISP-02**	Projector lamp brightness degraded	Optics cleaned and lamp timer reset	AV Vendor Technical Support	**Resolved & Verified**
2026-08-08	**GVEI-ICT-PC-078**	Network adapter disabled in BIOS	CMOS battery replaced & BIOS reconfigured	Internal Lab Administrator	**Resolved & Verified**
2026-09-12	**GVEI-PWR-UPS-01**	Battery bank impedance alert	Preventive battery terminal servicing	Power Maintenance Engineer	**Resolved & Verified**

Maintenance Classification Protocols

- **Tier 1: Preventive Maintenance:** Scheduled bi-weekly dusting, system health diagnostics, disk cleanups, and cable checks.
- **Tier 2: Minor In-House Servicing:** Input peripheral replacements, loose cable reseating, and software configuration fixes.
- **Tier 3: Specialized Support / Escalation:** Component repairs, motherboard/power supply replacements requiring external warranty support.

- **Tier 4: Service Testing Verification:** No equipment is returned to active classroom use until practical functionality has been verified by the HOD or Lab Assistant.

4. Software and Digital Learning Resources

The department audits software compatibility, licensing, and offline readiness to guarantee seamless laboratory practicals aligned with the approved CBSE/NCERT computing curriculum:

Software / Resource	Educational Purpose	Classes Served	Lab Availability	Licence / Access Status	Technical Action Required
Python IDLE & VS Code	Text-based coding, algorithms, CBSE projects	Class IX – XII	All Senior Workstations	**Open Source / Freeware**	Maintain updated standard runtime 3.11+
MySQL Server & Workbench	Relational database modeling, SQL queries	Class XI – XII	Senior Lab Network	**GPL / Community Edition**	Ensure auto-start database service on boots
Pandas, Matplotlib, NumPy	Data analysis, data visualization practicals	Class XII (IP & CS)	Senior Lab Systems	**Open Source Libraries**	Verify package versions via pip environment
Scratch 3.0 (Offline Editor)	Block-based coding, logic, game design	Class VI – VIII	Junior & Middle Labs	**Freeware / Educational**	Pre-install offline desktop runtime on all PCs
LibreOffice & MS Office	Spreadsheets, word processing, presentation	Class III – X	All Lab Systems	**Institutional Campus License**	Ensure offline template access for students
Tux Paint & Paint 3D	Foundational digital art & mouse fluency	Class I – V	Primary Wing Labs	**Freeware / Educational**	Pre-configure dedicated student export folders
GVEI LEADS / LMS Portal	Digital courseware, CBTs, resource sharing	Class I – XII	Campus LAN & Web	**Institutional Cloud Access**	Monitor bandwidth allocation during peak hours

Policy Standard: Software registers distinguish between currently installed, authorized applications and planned acquisitions. Offline desktop environments are prioritized to prevent instructional disruption during network interruptions.

5. Laboratory Timetable and Utilisation Framework

Laboratory timetables ensure equitable machine access, maximizing active hands-on student time across all school wings:

Date	Laboratory Wing	Class / Section	Period	Scheduled Learning Activity	Conducting Educator	Utilisation & System Remarks
2026-09-08	Senior ICT Lab	**Class XI-CS**	Period 3	Python Iteration & Looping Practical	Mr. Mehraj U Din Mir	100% workstation allocation; zero faults
2026-09-08	Middle ICT Lab	**Class VIII-A**	Period 4	Spreadsheet IF Function Drill	Assigned Faculty	38 students active; all exercises saved
2026-09-09	Senior ICT Lab	**Class XII-IP**	Period 2	MySQL Table Joins & Aggregations	Mr. Mehraj U Din Mir	Database server optimal; 100% pass

2026-09-09	Junior ICT Lab	**Class V-B**	Period 5	Tux Paint Creative Backgrounds	Assigned Faculty	Mouse fluency drills; outputs exported
2026-09-10	Senior ICT Lab	**Class X-IT**	Period 1	Digital Documentation & Stylesheets	Assigned Faculty	CBSE task sheet completed by all
2026-09-10	Middle ICT Lab	**Class VII-C**	Period 6	Scratch Event Broadcasting & Loops	Assigned Faculty	Interactive games tested in pairs

Laboratory Utilisation Audit Criteria

- **Curricular Adherence:** Confirming scheduled laboratory practicals execute as allocated in the school master timetable.
- **Machine Ratio Optimization:** Maintaining an optimal 1:1 student-to-computer ratio during core programming practicals.
- **Active Practice Pacing:** Ensuring students spend at least 70% of laboratory period time actively coding or executing tasks.
- **Conflict Avoidance:** Balancing routine classes, AI Club sessions, and Computer-Based Tests (CBT) without timetable clashes.

6. Technical Faults, Corrective Actions and Pending Issues

Technical challenges are documented through a central issue register to maintain full operational accountability and ensure rapid escalation:

Date Reported	Issue / Platform Affected	Instructional Impact	Corrective Action Deployed	Responsible Personnel	Current Status
2026-05-14	**LEADS Tab Slow Loading**	10-minute delay in launching online worksheet	Cache flushed; localized media mirror utilized	IT Systems Coordinator	**Resolved & Verified**
2026-06-20	**Senior Lab LAN Switch Drop**	Temporary loss of MySQL client connections	Patched uplink switch & secured cable housing	Network Administrator	**Resolved & Verified**
2026-07-08	**Video Streaming Buffering**	Lag during classroom AI demo video stream	Bandwidth QoS prioritized for Lab VLAN subnet	ISP Support / Network Team	**Resolved & Verified**
2026-08-16	**Python Interpreter Path Error**	Class XI workstations failed script execution	Reconfigured system environment variables globally	Internal Lab Administrator	**Resolved & Verified**
2026-09-02	**Projector HDMI Handshake Loss**	AV display flickered during Class IX lesson	Replaced active HDMI booster transmitter	AV Technical Assistant	**Resolved & Verified**
2026-10-04	**Secondary UPS Battery Warning**	Audible alarm during morning power transition	Preventive battery impedance calibration scheduled	Vendor Engineer / HOD ICT	**Action in Progress**

7. Internet Connectivity and Digital-Platform Reliability

Continuous campus network connectivity and learning management system reliability directly impact instructional momentum. When network or cloud-platform anomalies arise, educators implement a 6-step reporting and mitigation workflow:

7.1 Incident Timestamp & Diagnosis: Documenting exact time, affected lab, subnet, and nature of disruption.

7.2 Immediate Pedagogical Fallback: Switching immediately to offline desktop software runtimes or local mirror drives.

7.3 IT Cell & ISP Escalation: Lodging priority tickets with internal network admins or ISP gateway support.

7.4 Bandwidth Re-allocation: Implementing Quality of Service (QoS) throttling on non-academic network traffic.

7.5 Resolution & Pacing Verification: Verifying latency, ping stability, and platform tab responsiveness prior to closure.

7.6 Incident Archival: Cataloging recurring patterns to inform long-term infrastructure provisioning.

> **TECHNICAL INTEGRITY DIRECTIVE**
> *Platform Incident Standard: Issues concerning digital learning systems (such as LEADS portal tabs or streaming buffering) must be documented with verified timestamps, concrete instructional impact, and resolution proof. Never log unresolved issues as resolved.*

8. ICT Integration in Classroom Teaching

Technology serves as an accelerator of pedagogical efficacy rather than an ornamental display. Integration is evaluated by how digital tools deepen conceptual inquiry, problem-solving, and student creation:

Date	Class / Level	Curriculum Learning Objective	ICT Resource / Platform	Student Active Learning Task	Verified Learning Output
2026-05-12	Class XII-CS	Construct recursive data structures in Python	VS Code & Python 3.11	Code recursive binary search & trace stack frames	Verified script files & runtime call trace logs
2026-06-18	Class X-IT	Design automated mail merge documents	LibreOffice Writer	Create personalized certificates linked to data source	Exported PDF certificates & template files
2026-07-20	Class IX-AI	Understand computer vision & feature extraction	Teachable Machine & Webcams	Train image classifier for object detection in pairs	Trained model links & evaluation accuracy sheets
2026-08-14	Class VIII-ICT	Formulate spreadsheet conditional financial models	Spreadsheets & Formulas	Construct budget model with nested IF logic and charts	Dynamic spreadsheet files & chart printouts
2026-09-15	Class VII-ICT	Implement event broadcast loops in game design	Scratch 3.0 Desktop	Program multi-sprite interactive dialogue and game mechanics	Archived .sb3 project files & peer review notes
2026-09-22	Class VI-ICT	Develop computational sequencing & decomposition	Unplugged Drills & Flowcharts	Deconstruct everyday processes into algorithmic logic	Completed flowchart task sheets & rubrics

9. Laboratory Rules, Equipment Use and Safety Arrangements

The ICT Department enforces stringent laboratory hygiene, electrical safety, workstation discipline, and cyber ethics across all facilities:

Safety & Compliance Audit Dimension	Verification Status	Observed Compliance & Operational Standard
Laboratory Rules & Conduct Guidelines	Verified Compliant	Prominently displayed in English/Urdu at lab entry; communicated during orientation.
Cable Management & Concealment	Verified Compliant	All power conduits, LAN runs, and power strips routed through enclosed floor raceways.

Defective Equipment Restriction	**Verified Compliant**	Faulty terminals tagged with yellow 'Under Service' markers and isolated from student power.
Software Access & Cyber Security	**Verified Compliant**	Local admin accounts restricted; firewall content-filtering active; external USB ports locked.
Laboratory Supervision Pacing	**Verified Compliant**	Faculty and lab assistants present 100% of the period; zero unmonitored machine access.
System Shutdown & Storage Protocol	**Verified Compliant**	Standardized orderly OS shutdown; peripheral alignment and dust covers placed at day end.
Electrical Safety & Fire Preparedness	**Verified Compliant**	Centralized emergency power cut-off switch tested; CO2 fire extinguishers inspected and current.

10. Photographic and Documentary Evidence Standards

Visual records corroborate laboratory readiness, physical layout, and active student computational engagement. The department maintains an indexed photographic repository adhering to institutional child privacy policies:

- **Workstation Layout & Ergonomics:** Wide-angle photographs documenting clean raceways, unobstructed walkways, and machine spacing.
- **Active Practical Sessions:** Photographic evidence of students coding in Python, debugging Scratch projects, and querying MySQL databases.
- **Safety & Compliance Displays:** Images of displayed laboratory rules, emergency cut-offs, fire extinguishers, and terminal status tags.
- **Infrastructure Enhancements:** Before-and-after documentation of newly upgraded hardware, network rack cleanups, and display setups.

11. Infrastructure Review and Improvement Planning

Periodic reviews evaluate infrastructure capacity against upcoming curriculum changes and growing enrollment:

Infrastructure Domain	Current Empirical Observation	Evidentiary Source	Priority	Recommended Action	Status
Hardware Availability	82 operational systems out of 85 across two labs.	Physical inventory logs	**High**	Procure 15 high-spec workstations for senior AI modules	**Budget Approved**
Equipment Condition	Displays, keyboards, and mice operating normally.	Maintenance logbook	**Medium**	Replace aging mechanical keyboards with ergonomic units	**In Progress**
Software & Compilers	All standard IDEs installed; Python 3.11 active.	Lab software audit	**High**	Deploy standardized virtualized Linux programming labs	**Under Review**
Laboratory Utilisation	Average 88% timetable occupancy during school hours.	Master usage registers	**Medium**	Schedule dedicated after-school innovation club slots	**Scheduled**
Network & Connectivity	100 Mbps fiber uplink; localized streaming delays.	Bandwidth traffic logs	**High**	Implement dual-WAN failover & dedicated lab VLAN QoS	**Action Initiated**
Laboratory Safety	Fire, electrical, and cable safety 100% compliant.	Safety audit checklists	**High**	Conduct bi-annual laboratory emergency drill for students	**Completed**

12. Departmental Monitoring Indicators (KPIs)

The operational readiness and pedagogical efficacy of ICT facilities are measured against eight verified benchmarks:

Performance Indicator	Institutional Evaluation Standard
Physical Inventory Verification	100% of all computer assets and network devices physically audited bi-annually.
Workstation Operational Uptime	Sustained operational availability above 96% across all student workstations.
Fault Resolution Efficiency	Minor hardware/software issues resolved within 24 hours; critical network faults within 4 hours.
Laboratory Timetable Fidelity	Minimum 95% execution rate of scheduled laboratory practical periods without cancellations.
Student-to-Device Pacing Ratio	Maintenance of a strict 1:1 student-to-machine ratio during core senior programming practicals.
Syllabus Software Currency	100% availability of approved compilers, database engines, and runtime libraries prior to term start.
Authentic ICT Integration Index	Documented machine-based learning outputs collected for 100% of prescribed curriculum units.
Safety & Environmental Compliance	Zero electrical, fire, or cable hazards logged on quarterly campus quality reviews.

13. Evidence File Index for the Academic Audit

To ensure end-to-end operational traceability, the department maintains ten primary documentary dossiers:

13.1 Verified Laboratory Asset Inventory Register: Master hardware ledgers, serial ranges, and verified physical audit sheets.

13.2 Hardware Maintenance & Service Dossiers: Inspection checklists, component replacement receipts, and testing sign-offs.

13.3 Software Licensing & Installation Registers: Authorized software lists, compiler versions, and licensing certificates.

13.4 Master Laboratory Timetables & Class Logs: Allocated practical slots, room booking schedules, and attendance records.

13.5 Technical Fault & Resolution Tracking Log: Live ticketing register recording timestamps, diagnoses, and verified fixes.

13.6 Network & Platform Incident Reports: Bandwidth audit sheets, ISP ticket records, and LEADS portal logs.

13.7 Authentic Classroom ICT Integration Dossier: Lesson plans, task sheets, and student digital files demonstrating applied learning.

13.8 Laboratory Safety & Compliance Inspection File: Bi-weekly safety checklists, fire safety certifications, and emergency procedures.

13.9 Photographic Repository & Layout Diagrams: Captioned photographs of lab infrastructure, safety signs, and active coding classes.

13.10 Infrastructure Review & Enhancement Plans: Periodic SWOT reports, equipment requisitions, and approved budget roadmaps.

14. Priority Evidence Sets for the Academic Audit

For academic audit inspections, the department maintains three benchmark evidence dossiers readily accessible:

ACADEMIC AUDIT PRIORITY EVIDENCE SETS

- *Evidence Set 1 — Laboratory Readiness: The authenticated master asset register, working-status percentage, and latest hardware inspection sign-off.*
- *Evidence Set 2 — Technical Accountability: The live fault register detailing a reported technical issue, its documented instructional impact, remedial action, and final verified resolution proof.*
- *Evidence Set 3 — Educational Impact: An authenticated laboratory timetable record linked directly to a daily lesson plan, active student coding task, and verified digital student artifact.*

15. Expected Outcome and Departmental Commitment

This policy framework guarantees a well-maintained, highly dependable, and safe computing environment across Green Valley Educational Institute. It ensures that digital equipment is accounted for, technical disruptions are addressed transparently, laboratory time supports hands-on computational mastery, and infrastructure investments are guided strictly by pedagogical evidence.

Departmental Commitment: The ICT Department commits to maintaining rigorous infrastructure records, resolving technical hurdles proactively, enforcing responsible laboratory conduct, and continually aligning digital facilities with curriculum objectives and student innovation.

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